Studying the Human Mind with AI

From Behavioral Analysis to Role Simulation

Handout and reading companion for the talk

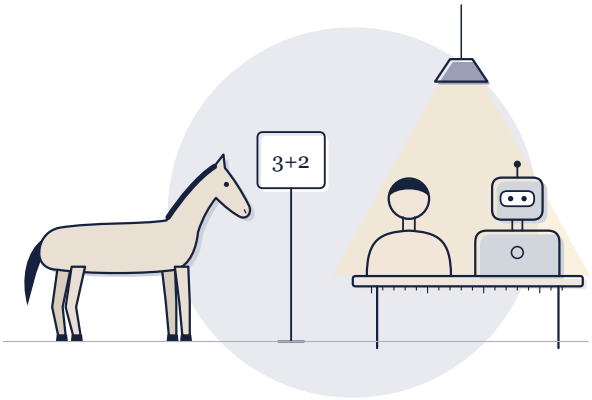
Shiyu Hu

Online:
https://github.com/huuuuusy/studying-the-mind-with-ai

Everything you need to start is in this handout, also if GitHub is slow or unreachable where you are.

scan for the repository

You heard the talk *Studying the Human Mind with AI: From Behavioral Analysis to Role Simulation*, or someone sent you this link. This page takes five minutes. It says what the talk was about, how this repository is organised, and where to go next.



The one question

It looks capable. How would we know it really is?

In September 2026, the US government told its agencies to stop saying ”artificial intelligence” and to say ”super intelligence” instead, because the technology is ”not simply imitating or automating human intelligence” (fact sheet[1]; order[2]). A new name does not settle what intelligence is. Even the best-known definition of superintelligence, ”an intellect that is much smarter than the best human brains in practically every field” (Bostrom, 1998, text[3]), is measured against people. Turing had stepped around the definition in 1950: he replaced ”Can machines think?” with a test against people, the Turing test (doi[4]). The talk opened with one real game from a recent run of that test: the judge took the AI for the human (card). Across all games, GPT-4.5 told to act like a shy young person was picked as the human 73% of the time; without that persona, 36%. It fooled the judges, and that shows how we judge, not that it can think. The test tells us about AI, and also about our own judgements.

Psychology is in the same position: it cannot open a mind, so it reads the mind from behaviour, against people. The older field and the younger one are natural partners, and AI has borrowed from psychology all along:

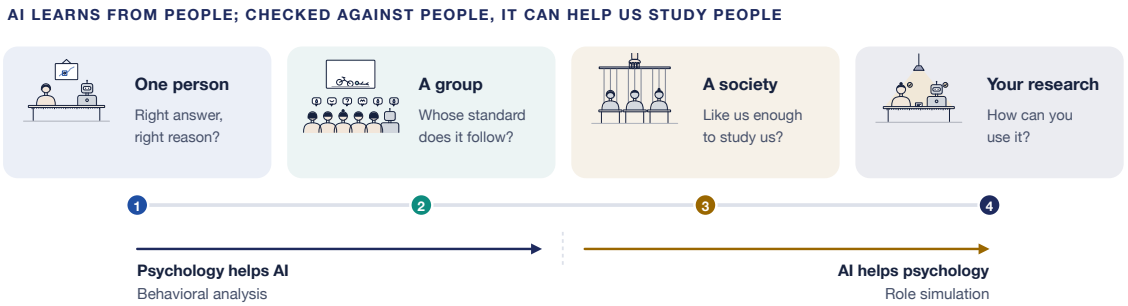
From psychology	Borrowed by AI
How a right answer can have a wrong reason: Clever Hans (1904)	shortcut learning (2020; Part 1)
How people see: a theory of human vision (Zhaoping, 2023)	the speaker’s group’s tracker (2024; Part 1)
How to judge: Thurstone’s paired comparisons (1927)	how ChatGPT learned what people prefer (2022; Part 2)
How we read events: Heider and Simmel’s moving shapes (1944)	scoring by a group of people, in the speaker’s group’s work (2024; Part 2)
How to be a person: the Big Five (Goldberg, 1990, doi[5])	the speaker’s group’s virtual students, with personas (2025; Part 3)

The direction can also go the other way: once a model passes the checks, it can give something back to the study of people.

[1] https://www.whitehouse.gov/fact-sheets/2026/09/fact-sheet-president-donald-j-trump-inaugurates-the-era-of-super-int
[2] https://www.whitehouse.gov/presidential-actions/2026/09/inaugurating-the-era-of-super-intelligence/
[3] https://nickbostrom.com/superintelligence
[4] https://doi.org/10.1093/mind/lix.236.433

The talk in one sentence: **AI learns from people; checked against people, it can help us study people.**

The talk in four parts



Four questions, each a little bigger than the last: one person, a group, a society, and your own research. Halfway, the direction changes: at first psychology helps us test AI (behavioral analysis, Parts 1–2); later, AI helps us study people (role simulation, Parts 3–4).

Part	The question	What the talk found	Guide
1 · One person: people next to the machine	Right answer, right reason?	Not yet: close scores, different mistakes. To check the reason, **compare the mistakes, not only the scores**.	Part 1
2 · A group: people as the answer	Whose standard does it follow?	A group of people, not one person. To check the standard, **compare with a group, not one person**.	Part 2
3 · A society: people we make	Is it like us enough to study us?	To rehearse a study, not to replace its people. To check a simulation, **compare it with real people**.	Part 3
4 · AI in your own research	How can you use it?	**Check it first, then use it**: when a model reads your data, test it on new people, look for the shortcut and keep the disagreement; when it takes a person's place, get the spread right, not only the average, and test its memory.	Part 4



Slide 48 of the talk: the whole talk on one page.

The checks in bold are ones psychology already has. Each test also taught us about people: how we see, how we understand events, and how we behave together. A question about AI keeps turning into a finding about us.

Each part's guide has the same sections:

- what the talk showed, usually in the talk's order: what psychology already knew, the classic work in AI, and what the speaker's group built;
- the question behind it;
- **what it gives back to psychology**;
- **five key readings**: what each did and found, what it means for the part, and what it does not show, each with a one-page card;
- **how they connect**;
- **the wider landscape**, with all the checked references for that part;
- the talk's own studies, in their public versions;
- the terms used;
- how to work on it with an AI researcher.

Why now

The talk closes with two things the speaker sees, and how the speaker would do research now ("Why psychology and the humanities matter more now").

- **Two stories from maths.** In May 2026 Timothy Gowers wrote that one AI proof could be "a perfectly reasonable chapter in a combinatorics PhD" (blog[6]). In July 2026 Claude found a counterexample to the 87-year-old Jacobian conjecture, for three or more dimensions, and it was checked in Lean (Tao[7]). In October 2026 OpenAI claimed results on hundreds of open problems (openai/math[8]), and three of those papers were withdrawn the next day. The speaker's reading: in some fields AI can now work like a PhD student, sometimes even like an expert, and every claim still needs checking.
- **Making is cheap; judging is scarce.** ICLR, one of the main AI conferences, had 490 submissions in 2017 and 19,525 in 2026, and about 62,000 have been submitted for 2027, not yet reviewed. At ICLR 2026, 18,054 reviewers wrote 76,139 reviews, and the programme chairs wrote that the scale "made it impossible to rely on human judgement alone to consistently flag and discover problematic reviews" (ICLR Blog[9]).

[6] https://gowers.wordpress.com/2026/05/08/a-recent-experience-with-chatgpt-5-5-pro/
[7] https://terrytao.wordpress.com/2026/07/21/a-digestion-of-the-jacobian-conjecture-counterexample/
[8] https://github.com/openai/math
[9] https://blog.iclr.cc/2026/03/31/a-retrospective-on-the-iclr-2026-review-process/

How the speaker would do research now:

- **Outlast the model.** Ask: would it still matter if models were ten times better?
- **Let AI do; you judge.** Hand it the doing, keep the judging, and write the first draft yourself.
- **Find your own crossing.** Start from a question your field asks best, and bring AI to it. For example: does talking to AI add to human contact, or replace it?

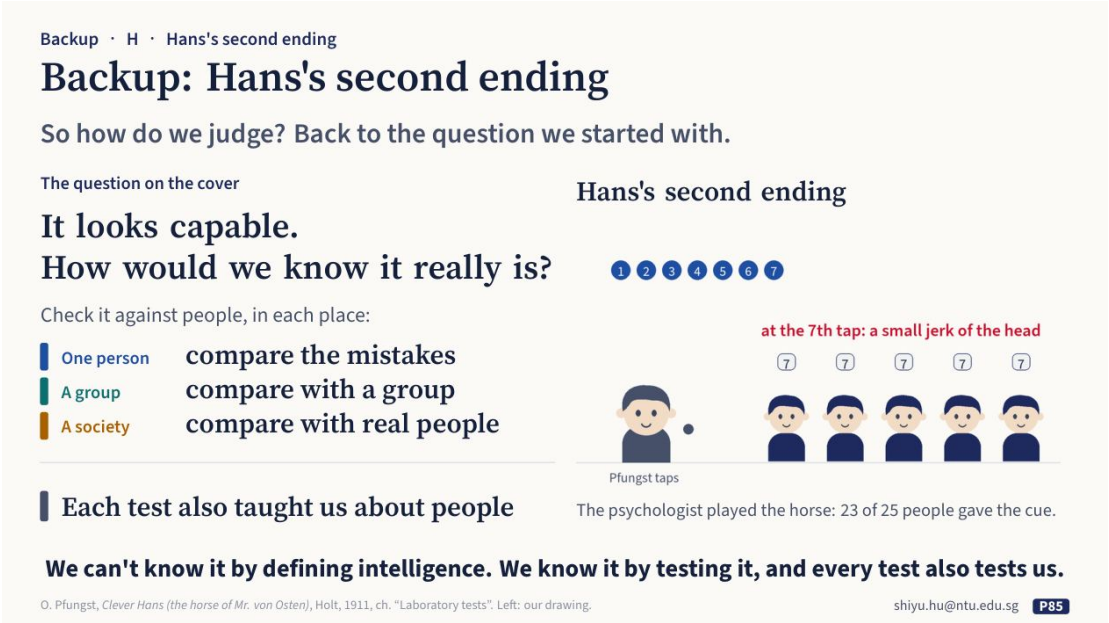
The talk's view: when making things gets cheap, judging is the real work, and that has always been the job of psychology and the humanities.

How the talk ended

The summary page gave each part's answer and came back to the line from the start: **AI learns from people. If we check it against people, it can help us study people.**

A backup page holds a second ending. It goes back to the question on the cover: how would we know a machine really has an ability? Not by giving it a new name, but by checking it against people: compare the mistakes, compare with a group, compare with real people.

Then it returned to Clever Hans. After he had solved the puzzle, Pfungst played Hans himself: he asked 25 people each to think of a number and tapped like Hans, and when he reached it, 23 of them gave a small nod without knowing it (Pfungst, 1911, Project Gutenberg[10]). A test of a horse became a finding about people, and the talk's tests do the same: they teach us how people see, understand events and act together. The page's last line: **we can't know it by defining intelligence; we know it by testing it, and every test also tests us.**

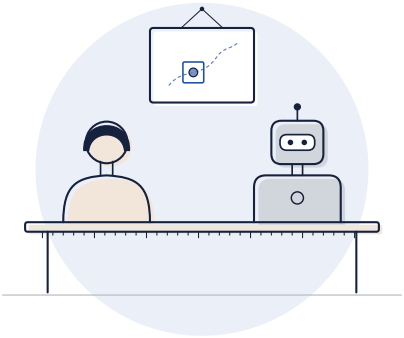


Backup slide 85 of the talk: Hans's second ending.

[10] https://www.gutenberg.org/ebooks/33936

Part 1 · One person: people next to the machine

Right answer, right reason? The check: ***compare the mistakes, not only the scores****.*

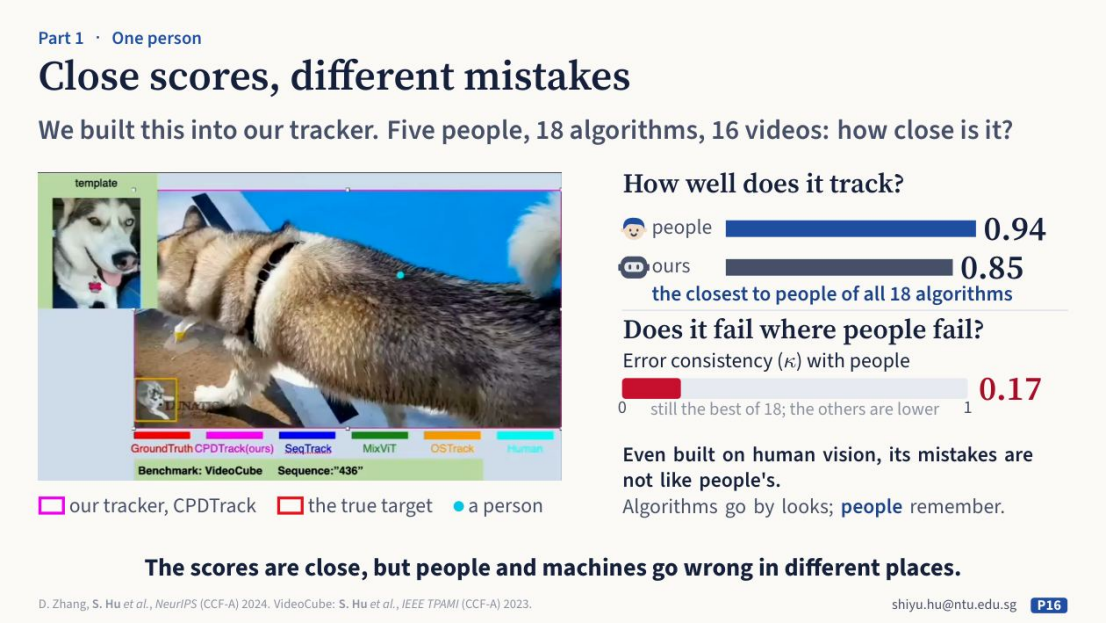


What the talk showed

What psychology already knew. Clever Hans, the horse who seemed to count, was tested by changing one thing: did the questioner know the answer? When the questioner knew, Hans was right 98% of the time. When the questioner did not know, he was right 8% of the time. Hans was reading tiny, unintended movements of the questioner's head, not doing arithmetic. Behaviour that looks like an ability is not yet evidence of it; a controlled comparison is.

The classic work in AI. AI has its own Clever Hans. Sebastian Lapuschkin and colleagues found a model that recognised horses by a source tag on the photos, not by the horse. Robert Geirhos and colleagues in Tübingen showed that networks trained on ImageNet (a large collection of labelled photos) call a cat with the skin of an elephant an elephant: they go by texture, people go by shape. Geirhos and colleagues named the general problem shortcut learning: a cue that works in training, not the ability we meant. A shortcut can still score well, so to catch one, compare the mistakes. Geirhos and colleagues measured this with Cohen's κ on which images each one gets wrong. The talk's small example: two models both get 7 of 10 images right. One fails on the same three images as a person ($\kappa = 1.00$); the other shares only one ($\kappa = 0.05$). The same score; only the mistakes tell them apart.

What the speaker's group built. Tracking, with people as the reference. Tracking means following one object through a video, even when it goes out of sight. AI's usual test is a toy example: one short, continuous shot, so a model need only search near the last spot. The group used a harder task, more like real video and discontinuous in time and space: the shot cuts, the target disappears, then comes back somewhere else. On it, people did better than twenty tracking algorithms. A clip from the group's human–machine study shows the difference. Among look-alike flamingos, a person stays on the same bird; a state-of-the-art algorithm jumps to another bird that looks similar. The person keeps to the target; the algorithm goes by looks. Then came a tracker built from how people find a target again, CPDTrack. In the theory it borrows, the periphery is for looking and the centre is for seeing: wide side vision finds where to look, and narrow central vision checks what is there. Five people and 18 algorithms tracked the same 16 videos. People scored 0.94, and CPDTrack scored 0.85, the closest to people of all 18. But its error consistency with people was only 0.17 (0 is chance, 1 the very same errors), and that was the highest of the 18. Even built on human vision, its mistakes are not like people's: algorithms go by looks; people remember.



Slide 16 of the talk: close scores, different mistakes.

The question behind it

Right answer, right reason? Not yet: close scores, different mistakes. Scores show the result; the process shows where people and machines go wrong. To compare a model with people, compare **which** items each gets wrong, not only how many, and put the cue you suspect in conflict with the one you intend: a controlled comparison, in psychology's terms. The human reference is the spread among people, not a ceiling of 100%.

To check the reason, compare the mistakes, not only the scores.

What it gives back to psychology

The tests also teach us about people, here about how we see. But a model built from a theory of human vision does not test that theory on its own. Taking one part of the tracker away shows what the model needs, not how people see. Whether the split between central and peripheral vision holds for people needs people: for example, masking central or peripheral vision and seeing whether the same kinds of error rise. A backup page of the talk leaves this as an open question: do a model's parts fail like ours?

Key readings

Read these five first. Each one below says what was done, what was found, what it means for this part's question, and what it does not show; its card gives the method in full and where the data are.

1. A cat with elephant skin: ImageNet-trained networks see texture, people see shape

Geirhos et al., 2019, *ICLR*

What they did. A cue-conflict design, the standard way perception research asks which of two cues an observer uses. Style transfer (a method that redraws one image in the texture of another) gave each image the shape of one category and the texture of another, and people and networks classified the same images. The human data come from laboratory psychophysics: nine experiments, 48,560 trials, 97 observers.

What they found. Networks trained on ImageNet, a large collection of labelled photos, are strongly biased towards texture, while people decide by shape. Trained on *Stylized-ImageNet*, whose textures have been replaced, the same network design (ResNet-50, a standard image network) learns a shape-based representation instead, fits human behaviour better and is more robust to image distortions.

What it means here. This is the talk's cat with elephant skin, and the plainest form of this part's question: put the suspected cue in conflict with the intended one and see which wins, for people and machines alike. The answer also depends on training data, not only on the architecture.

What it does not show. Shape versus texture is one cue dimension among many. People and a model can agree on which cue wins and still fail on different images; reading 2 measures that.

2. Close scores, different mistakes: error consistency

Geirhos, Meding & Wichmann, 2020, *NeurIPS*

What they did. For every trial they recorded whether each decision maker was right or wrong, and compared how often two of them are wrong on the same trials with how often chance would predict from their accuracies. The index is Cohen's κ on right and wrong, a statistic from psychological measurement, usable for model–model, person–person and model–person pairs.

What they found. On object recognition, networks of any architecture are remarkably consistent with one another, but their consistency with human observers is little above chance: on one set of images, 0.33 between people and only 0.07 between networks and people. CORnet-S, a model of the brain's object-recognition pathway with loops that feed signals back (recurrent), behaves here essentially like a standard image network in which signals pass only one way (feedforward ResNet-50).

What it means here. This is this part's check in its exact form: compare which items each system gets wrong, not only how many. It goes one step beyond reading 1, from which cue wins to whether failures fall on the same images, and turns "the model processes images like people do" into a test that can fail.

What it does not show. Consistent errors are necessary for a shared strategy but not sufficient, and a low κ alone does not identify which shortcut the model uses. Read κ against the consistency among people, not against an ideal of 1.

3. A right answer found in the wrong place: Clever Hans predictors

Lapuschkin et al., 2019, *Nature Communications*

What they did. Explanation methods show which parts of an input a model relied on, and the paper's Spectral Relevance Analysis (SpRAy) groups these maps across many inputs, so that whole strategies show up, not single cases. The decisive test is the one Oskar Pfungst used on Clever Hans in 1904: remove the suspected cue and see whether the answer survives, then add it where it does not belong and see whether the answer follows.

What they found. An image classifier (a Fisher vector classifier, an older, pre-deep-learning kind) trained on PASCAL VOC 2007, a public photo collection, recognised horses by a source tag that appears in about one-fifth of the horse images. Removing the tag removed the horse classification; adding it to a car image changed that image's classification.

What it means here. This is AI's own Clever Hans from the talk, and its logic is psychology's: a manipulation that removes the cue is a control condition. Geirhos and colleagues later named the general problem shortcut learning. Reading 2 can show that a model differs from people; this shows how to pin down the cue it uses instead.

What it does not show. Explanation maps are measurements that need their own validation: they show where the model looked, not why. The removal and insertion test is what makes the conclusion convincing.

4. Partial success: the robustness gap closes, the consistency gap does not

Geirhos et al., 2021, *NeurIPS*

What they did. People were tested on out-of-distribution data sets, images unlike those the models were trained on, such as noisy, stylised or edge-only images: 85,120 psychophysical trials across 90 participants. The models differed from standard networks trained on labelled photos in what they were trained to do, in their design and in how many images they learned from (1 million to 1 billion), and each was compared with people on accuracy and on image-level error consistency.

What they found. The robustness gap is closing: on most data sets the best models now exceed human performance under brief presentation. A substantial image-level consistency gap remains, and most models agree with one another in their errors. In many cases, consistency with people improves when the training set grows 10 to 1,000 times larger.

What it means here. Not shown in the talk. Later, the best models matched people's accuracy, but not their errors. It shows why accuracy and error consistency must be reported separately: whether a model is "more human-like" depends on which one you measure. The human trial data are public.

What it does not show. The human data come from brief presentation of single images, so they describe fast recognition, not everything human vision does. The results describe the models tested in 2021.

5. Seeing but not perceiving: BLINK

Fu et al., 2024, *ECCV* (BLINK)

What they did. Fourteen classic computer vision tasks, such as relative depth, visual correspondence and forensics detection, were reformatted into 3,807 multiple-choice questions with one or more images, put to multimodal language models (chat models that take images as well as text). Human accuracy on the same questions is the baseline; random guessing gives a second reference point.

What they found. People reach 95.70% accuracy on average. The best models at the time, GPT-4V and Gemini, reach 51.26% and 45.72%, only 13.17 and 7.63 percentage points above random guessing. Specialist computer vision models do much better.

What it means here. Not shown in the talk, it carries this part's question from image classifiers to the chat-style models that most people now use: fluent answers can hide weak perception. Read it last, for how far the question reaches today.

What it does not show. It compares accuracy only, not which items people and models fail on, the step reading 2 would add. The results describe the models tested in 2024; newer models should be tested again.

How they connect

Three threads run through these readings.

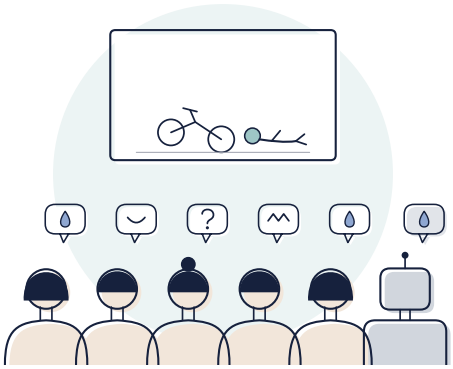
- **The cue.** Pfungst showed in 1904 that Clever Hans answered from tiny, unintended movements of his questioner's head, by removing that cue and watching the answers fail. Lapuschkin et al. applied the same logic to a classifier. Geirhos et al. (2020, *Shortcut learning in deep neural networks*, doi[11]) then gave the general name: shortcuts are decision rules that work on standard benchmarks but fail under harder test conditions. That paper notes that related problems are known in comparative psychology, education and linguistics.
- **The measurement.** The texture-bias study asks which of two cues wins. Error consistency goes one step further: do people and models fail on the *same* images? It borrows Cohen's κ, a statistic from psychological measurement. Partial success applies both to many newer models and publishes the human trials.
- **The scope.** The same comparison now reaches multimodal language models. BLINK reports accuracy against a human baseline; it does not yet report which items people and models fail on, which is the natural next step.

The open question. *Partial success* finds that models now match or beat people on robustness while still failing on different images. Whether a model is "more human-like" depends on which of the two you measure, so report both.

[11] https://doi.org/10.1038/s42256-020-00257-z

Part 2 · A group: people as the answer

Whose standard does it follow? The check: ***compare with a group, not one person****.*



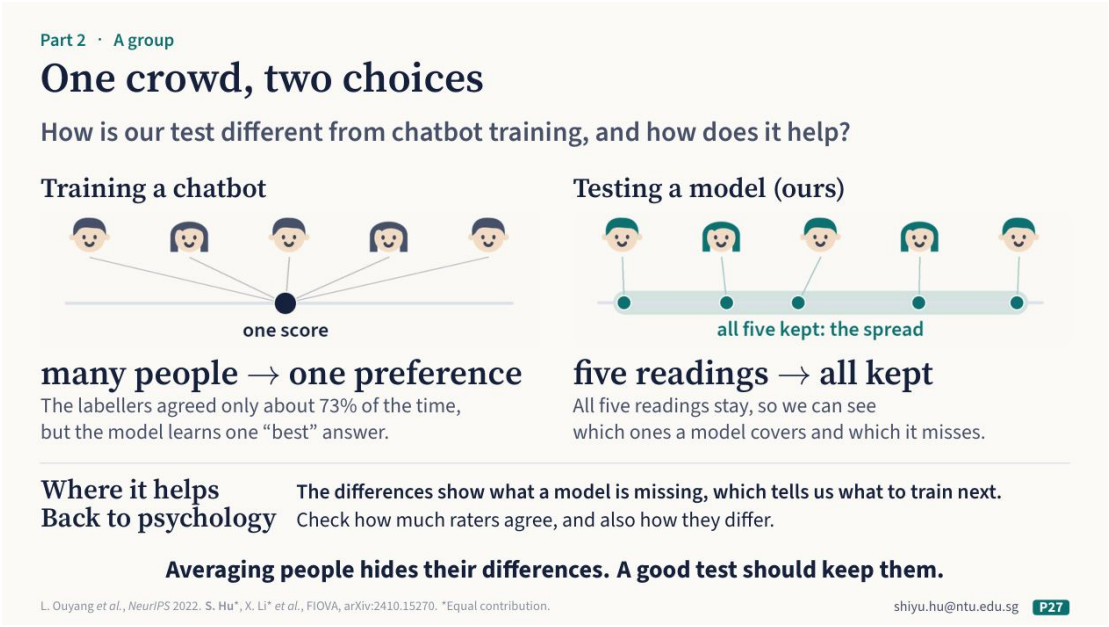
What the talk showed

What psychology already knew. Shown moving shapes, people read intentions into them, and each reads them a little differently: people are not one ruler but a spread. And Thurstone showed in 1927 how to turn many people's choices between pairs into one scale.

The classic work in AI. Just before ChatGPT, this idea from psychology taught models what people want. In 2020, GPT-3 had 175 billion parameters (the numbers a model adjusts as it learns). Size alone did not give answers people liked. InstructGPT, in March 2022, used reinforcement learning from human feedback (RLHF): people ranked several answers to the same prompt, and each ranking was split into pairs, paired comparison again. A scoring model (a "reward model") was trained on the pairs, and the chat model learned to score well on it. People then preferred the answers of a 1.3-billion-parameter InstructGPT model, more than a hundred times smaller than GPT-3. ChatGPT, that November, used the same method. A backup page adds that chatbots are ranked from people's paired votes too. But the rankings came from about 40 labellers (raters, in psychology's terms), and they agreed with each other only about 73% of the time. Whose preferences did it learn?

What the speaker's group built. Understanding video, with people as the reference. Even those labellers did not always agree: people's judgements are a spread, not one answer. AI's usual test misses that. It asks for one sentence and scores it against one reference, like a final score. So the group took understanding apart, the same idea as in Part 1: do not stop at the final score, look at the process. What happened, from an action to the full story. Why, step by step, with CausalStep. Describing comes close to people: 92.0% right for people, 85.2% for the best model. Explaining why does not: 76.8% for people, 39.8% for the best model, about half. And whose reading, with FIOVA, in which five people describe each of 3,002 clips. In one, a child on a balance bike falls. All five agree that he ends up on the ground. They do not agree on what it means: one says he is pretending, others that he cries or smiles. An AI model picked one reading, crying, and stated it as a fact. So all five descriptions are kept, and a model is scored against the group, event by event. This cognitive weighting gives an event more weight the more the five stress it; that is agreement among viewers, not proof that it happened. One model's description covered about a third of the weight. Across about 3,000 clips, where people disagree most, models score lower and look more alike.

One crowd, two choices. Training a chatbot averages people into one preference, so the model learns one "best" answer, though the labellers agreed only about 73% of the time. The group's test keeps all five readings, so we can see which ones a model covers and which it misses, and that shows what to train next. Averaging people hides their differences. A good test should keep them.



Slide 27 of the talk: one crowd, two choices.

The question behind it

Whose standard does it follow? A group of people, not one person. Preference training ends in one reward score; an evaluation can keep the spread. One label, or a majority vote, hides how much people disagree. Treat the model as one more rater: does it agree with people as well as they agree with each other, and does it keep their disagreement or flatten it? In psychology's terms: inter-rater reliability, with the spread among raters as part of the data.

To check the standard, compare with a group, not one person.

What it gives back to psychology

Psychologists now use language models to code answers and rate material. You already check human raters by how much they agree; keep how they differ, too. Before a model codes or rates for you, check it against the spread of a group, not against one answer key. The speaker's video data, on a backup page of the talk, show why. On the clips where the five people differed most, all six models scored lower. The models also looked more alike: how much a clip's rank varied across the six models fell from 0.577 on the clips where people agreed most to 0.444 on the 169 clips where they differed most, almost a quarter less. Where people spread out, the models moved closer together. And the data run the other way as well: thousands of clips with five readings each show where and how people's readings of an event differ.

Key readings

Read these five first. Each one below says what was done, what was found, what it means for this part's question, and what it does not show; its card gives the method in full and where the data are.

1. People read intentions into moving shapes

Heider & Simmel, 1944, *American Journal of Psychology*

What they did. A film of about two and a half minutes showed three geometric figures moving in and around a rectangle with a door. Three groups of undergraduate women, 114 in all, watched it twice: one was only asked to write down what happened, one to interpret the movements as actions of persons, and one saw the film in reverse.

What they found. Almost every participant described the film as actions of animated beings, mostly persons. Some readings were almost uniform: every participant saw the figures pushing or pulling the door, forwards or in reverse. Readings of motives and causes varied much more, especially for the reversed film.

What it means here. This is the classic behind the talk's moving shapes: observers agree on actions and diverge on meanings. The five people who describe the child falling off a balance bike show the same split between seen and inferred, and reading 4 finds it in emotion ratings. The spread of a group belongs in the reference, and it is widest on

what is inferred.

What it does not show. The participants were 114 undergraduate women in the 1940s, and the study reports descriptions and their themes, not a modern agreement statistic.

2. A chat model's training rests on people's comparisons

Ouyang et al., 2022, *NeurIPS*

What they did. Labellers (raters, in psychology's terms) wrote demonstrations of the desired behaviour for a set of prompts, and GPT-3 was fine-tuned on them, that is, trained a little further on these examples. Labellers then ranked several outputs for the same prompt. The rankings trained a reward model, the scoring model that gives each answer one score. The language model was then trained to raise that score, a method called reinforcement learning from human feedback (RLHF). The resulting InstructGPT models were evaluated by asking people which outputs they preferred.

What they found. On the authors' prompts, people preferred a 1.3-billion-parameter InstructGPT model to the 175-billion-parameter GPT-3, which has over 100 times more parameters (the numbers a model adjusts as it learns). The labellers who made the training data agreed with each other about 73% of the time (72.6 ± 1.5%).

What it means here. This is the talk's classic work in AI: the reference behind today's chat models is a group of people making comparisons, and they disagree about a quarter of the time. Training ends in one score from the reward model, so their spread is not kept. Whose judgements count, and what happens to their disagreement, is a measurement question.

What it does not show. The model is aligned to, that is, made to follow, one particular group, the authors' labellers, mostly English-speaking people living in the United States or Southeast Asia, and to the researchers' own instructions; not to people in general.

3. About fifty people per image: CIFAR-10H

Peterson et al., 2019, *ICCV* (CIFAR-10H)

What they did. People categorised every image in the CIFAR-10 test set, 10,000 small images in ten classes. In all, 2,571 participants on Amazon Mechanical Turk each categorised 200 images, with an attention check every 20 trials. Classifiers were then compared with the resulting label distributions and also trained on them.

What they found. The 511,400 human decisions, about 50 per image, give a full distribution of labels for each image. Classifiers trained in the usual way do not show human-like uncertainty. Training on the human distributions closes this gap. The models then do better on test images increasingly unlike their training images, and are harder to fool with adversarial attacks (images changed slightly on purpose to make a model err).

What it means here. Not shown in the talk. It turns disagreement from noise into data: the same image can be a cat to most people and a dog to some. It is the simplest case of this part's check, a group rather than one label, and it shows that the comparison changes what a model learns. Read it before reading 5, which surveys its research area.

What it does not show. The images are tiny and the ten categories simple, and crowd workers on one platform are one particular human reference.

4. Forty-four raters, fourteen films: Emo-FilM

Morgenroth et al., 2025, *Scientific Data* (Emo-FilM)

What they did. Forty-four raters annotated 14 short films, more than two and a half hours in total, on 50 items: discrete emotions, plus components of emotion from appraisal, motivation, motor expression, physiological response and feeling. A separate sample of 30 people watched the same films during functional MRI, with heart rate, respiration and electrodermal activity recorded.

What they found. The ratings had a mean inter-rater agreement of 0.38: moderate at best, although every rater saw the same films.

What it means here. Not shown in the talk. This is Heider and Simmel's split in modern emotion ratings: when raters judge feelings, agreement is only moderate. A model that rates emotion in these films should be judged against this spread, not against one rater or the average rater. As this part's open question puts it, at 0.38 "agreeing with the human label" is not a well-defined target.

What it does not show. The data were designed for affective neuroscience, and 0.38 is a mean: agreement on single items and moments can be higher or lower.

5. When people disagree, learn from the disagreement

Uma et al., 2021, *JAIR*

What they did. A survey with its own experiments. It reviews the evidence for disagreement in language and vision tasks, from objective ones such as part-of-speech tagging (marking each word as a noun, verb and so on) to subjective ones, then trains the most common approaches on every available data set with multiple judgements per item.

What they found. The ranking of training methods depends on the evaluation chosen, so the field must agree on how to evaluate even without a single gold standard. Soft labels keep the share of people who chose each answer; a gold label keeps one answer. With many judgements by high-quality coders for each item, training directly on soft labels beat training on combined or even gold labels. This held both when models were scored against one answer (hard evaluation) and against the spread of answers (soft evaluation). Otherwise, combining gold and soft labels generally did best in hard evaluation.

What it means here. Not shown in the talk. It is the entry point to the research area that treats the spread among people as information, and a map of its data sets. It answers this part's question from the side of machine learning: one label, or a majority vote, hides how much people disagree. Read it after reading 3.

What it does not show. It covers language and vision tasks up to 2021; large language models as raters came later (see the wider landscape).

How they connect

Four threads run through these readings.

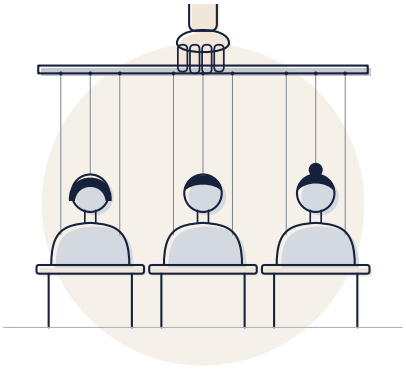
- **Agreement as measurement.** Psychology has long measured how far raters agree: Cohen's κ for two raters and categories (1960), and Krippendorff's α for any number of raters and kinds of data (Hayes & Krippendorff, 2007, doi[12]). The habit in this part treats a model as one more rater and asks whether it agrees with people as well as they agree with each other.
- **Disagreement as data.** Dawid and Skene (1979, doi[13]) estimated each observer's error rates instead of trusting a majority. Aroyo and Welty (2015, *Truth is a lie*, doi[14]) argued that there is often no single true label to recover. CIFAR-10H collected the full human distribution for an image data set, and Uma et al. brought the field together.
- **Seen and inferred.** Heider and Simmel found near-uniform readings of actions and varied readings of motives. Emo-FilM shows the same pattern in modern emotion ratings: agreement is only moderate when raters judge feelings.
- **Human judgement as training signal, and its replacement.** Thurstone turned people's choices between pairs into a scale (1927, *Psychological Review*, doi[15]); Bradley and Terry gave the model that today's reward models use (1952, *Biometrika*, doi[16]). Learning from human preferences (Christiano et al., 2017, NeurIPS[17]) became the training method behind InstructGPT. Language models are now used as judges in place of people. Zheng et al. (2023, *Judging LLM-as-a-judge*, doi[18]) report over 80% agreement between GPT-4 and human preferences, the same level as between humans, and also position, verbosity and self-enhancement biases.

The open question. If people agree only 73% of the time on which answer is better, or at 0.38 on emotion, then "agreeing with the human label" is not a well-defined target. A model that agrees with the majority more than any person does may have *flattened* the disagreement rather than matched it. Report the model's agreement against the range of agreement among people, and whether it keeps their spread.

[12] https://doi.org/10.1080/19312450709336664
[13] https://doi.org/10.2307/2346806
[14] https://doi.org/10.1609/aimag.v36i1.2564
[15] https://doi.org/10.1037/h0070288
[16] https://doi.org/10.2307/2334029
[17] https://papers.nips.cc/paper/2017/hash/d5e2c0adad503c91f91df240d0cd4e49-Abstract.html
[18] https://doi.org/10.52202/075280-2020

Part 3 · A society: people we make

Is it like us enough to study us? The check: ***compare it with real people****.*



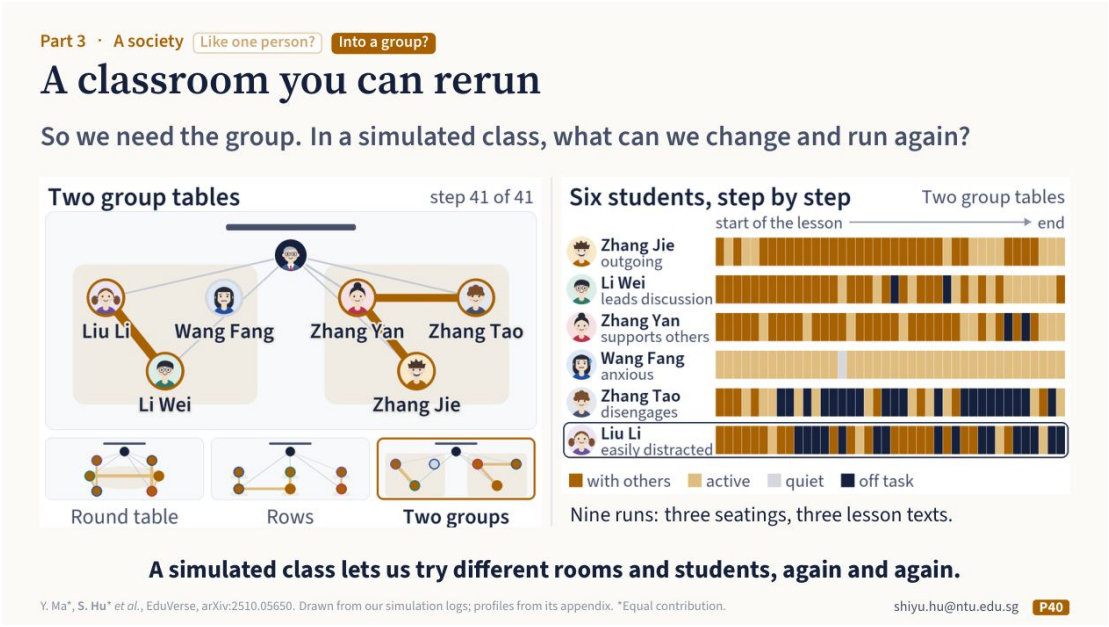
What the talk showed

The classic work in AI. Smallville grew out of a design need. In human–computer interaction, a new online community is tried out by a small group, but trolls and fights show up only when thousands join. So in 2022 a team let a language model play the users, troublemakers included. In 2023 the same team built Smallville: twenty-five agents with memory, reflection and plans. Take those parts away, and people rate the agents as less believable. One invitation to a party then spread through the town: about a dozen were invited and five came, and the talk among them was not scripted. It was a big step, with limits: the agents rarely said no, spoke too formally and walked into an occupied bathroom, and the authors warn that they should never replace real human input. Seeming real is not yet being real. Then the team interviewed 1,052 people and built a digital twin of each. To test a copy of a person, compare it with that same person, asked twice. Two weeks apart, people matched their own answers about 80% of the time; the twins matched them about two-thirds of the time (65.7%), or 0.83 of people's own consistency.

What psychology already knew. Psychology maps personality with five broad traits, the Big Five (Goldberg, 1990, doi[19]). A classroom has two things a town does not: established theory like this, and real classes to compare with. The talk's backup pages add a check of their own. A persona is a condition the experimenter sets, so psychology treats it as a manipulation: set it, measure it, and check that it moved the trait it names, and only that trait. Another lab has run such a check (reading 5): each trait was prompted on its own, from low to high, and mainly the trait that was prompted moved. So the check can pass, but it has to be run on the model you use.

What the speaker's group built. Virtual students, with people as the reference. Two questions: is one student like a real student, and does a whole class work like a real class? There are two routes to a simulated student. Writing a persona into a prompt for a closed model is the easier one; the group took the other, fine-tuning an open-source model. In the first study, EduEval, a model wrote classroom replies for each Big Five persona, starting from real lesson transcripts, lesson plans and textbooks. Teachers corrected the replies, and their corrections became the examples for the rest. A small add-on module was then trained on this material for each persona; the large model underneath was not retrained. Ten raters with experience of teaching Chinese judged replies without knowing which came from AI, and after training they rated the AI replies about as real as real students' replies. A reply can look real without a mind behind it. And one reply is not enough: ten trainee teachers each taught five virtual students, in 50 sessions and 1,288 turns. The persona mostly held, but one type, low conscientiousness, changed a lot; whether real students vary like that still has to be checked. Then a whole simulated class, EduVerse, where each student reacts to the teacher, the room and the classmates nearby. The class was checked against one classic pattern of real classroom talk (the teacher asks, a student answers, the teacher gives feedback): close for one kind of text, different for two others. Two simulated students stayed very different across nine lessons: the outgoing one joined in at 52% of the steps, the anxious one at 1%, though her profile says she gradually joins in. Only a group shows this. And a simulated class can be rerun: nine runs, with three seatings and three lesson texts, changed who talked to whom.

[19] https://doi.org/10.1037/0022-3514.59.6.1216



Slide 40 of the talk: a classroom you can rerun.

The question behind it

Is it like us enough to study us? Enough to rehearse a study, not to replace its people, as long as you start from real people and check every step. Acting like a kind of person is not yet being like one. A persona is a condition you set, so it needs a manipulation check: did it move the trait it names, only that trait, and under every wording that should mean the same? A blind rating shows whether answers *look* human, not whether a mind lies behind them. A simulation is useful once it is compared with real people, at the level the claim is about: one person, a group, or a whole class.

To check a simulation, compare it with real people.

What it gives back to psychology

Here the direction of the talk changes. In the first half (behavioral analysis), psychology helps us test AI; from here on (role simulation), AI helps us study people. A checked simulation is a new tool for psychology and the social sciences:

- **Many different participants, on demand.** Simulated people can rehearse a study before it reaches real ones: try the materials, the instructions and the design first.
- **A class you can rerun.** The same students and the same lesson, with the seating changed or one student swapped, run again and again: something no real school can offer. Try it in the simulation first, then confirm it with real people.
- **How to check one.** Start from real people; judge a simulated person against the same person measured twice; compare the group's spread with people's, not only its average; and watch for personas that never change and agents that, as in Smallville, rarely say no.

Key readings

Read these five first. Each one below says what was done, what was found, what it means for this part's question, and what it does not show; its card gives the method in full and where the data are.

1. Social simulacra: a community populated before it opens

Park et al., 2022, *UIST*

What they did. A designer describes a new online community in plain language: its goal, its rules and a few member personas. A chain of prompts to GPT-3, each building on the answer to the last, expands this into thousands of members who post and reply, including anti-social behaviour. The authors regenerated 50 real subreddits (discussion groups on Reddit) from their goals and rules alone, and asked 200 participants to pick the real conversation from pairs of one real and one generated.

What they found. Participants misidentified on average 41% (SD = 10) of the pairs; guessing at random gives 50%. Conversations written by crowd workers were easier to spot (32%).

What it means here. This is where Smallville came from: a designer needed many different test users, on demand, the first thing this part says a checked simulation gives back. It also set an early criterion for simulated people: can people tell generated behaviour from real behaviour? Readings 2 and 3 show the same team moving from there towards comparison with real people.

What it does not show. Telling real from generated tests plausibility, not accuracy: a thread can look real without matching how that community would develop. The evaluation used Reddit, English, U.S. participants and the 2020 GPT-3 model, before it was trained to follow instructions.

2. A town of twenty-five generative agents

Park et al., 2023, *UIST*

What they did. Each agent records its experiences in natural language, condenses them into higher-level reflections, and retrieves them to plan what to do next. Twenty-five agents live in a small simulated town, like a game world, inspired by *The Sims*. A hundred people rated how believable the agents were. The authors also built reduced versions, taking the parts away one after another: first reflection, then planning as well, then memory too.

What they found. The agents produced believable individual and emergent social behaviour. From one user-specified idea, that one agent wants to throw a Valentine's Day party, the agents spread invitations, asked each other out and arrived together at the right time. Each part of the mind made the agents more believable. The full agents scored 29.9; without reflection, 26.9; without reflection or planning, 25.6; with no memory, reflection or planning, 21.2. People playing the same roles scored 23, below the full agents.

What it means here. This is the study behind Smallville in the talk, and the one that made simulated people with language models a field. Its criterion is believability, which is not yet the same as behaving like real people. As this part notes, its agents also rarely say no, a sign to watch for in any simulation.

What it does not show. The study shows that the behaviour looks human to observers; it does not compare the agents with how real people in a real town behave. Reading 3 is the same team's answer to that gap.

3. One agent for each of 1,052 people

Park et al., arXiv:2411.10109

What they did. A stratified sample of 1,052 Americans each gave a two-hour interview to an AI interviewer, then completed the core General Social Survey (GSS), the Big Five Inventory, economic games and published experiments, twice, two weeks apart. Agents built on GPT-4o were given each person's interview, survey answers or both, and compared with agents given only demographics. Normalised accuracy divides how often an agent matches its person by how often the person matches their own earlier answers.

What they found. On GSS items that were not part of what the agents were given, interview agents gave their person's answer 65.67% of the time; people repeated their own answer 79.53% of the time: a normalised accuracy of 0.83 (current version, v3). Demographics alone reached 0.74.

What it means here. This is the step in the talk from invented residents to real people. A simulated person is judged not by whether it looks human, as in reading 2, but against the same person, measured twice. People's own two-week consistency becomes the ceiling: this part's check, made exact for one person.

What it does not show. Matching individuals is not reproducing how attitudes go together across a sample. In the economic games, no agent type significantly beat demographics alone, and in the experiments agents' effect sizes were larger than people's.

4. Out of one, many: silicon samples

Argyle et al., 2023, *Political Analysis*

What they did. The authors gave GPT-3 thousands of sociodemographic backstories taken from real participants in large U.S. surveys, and asked it to answer as each person would (*conditioning* the model, in AI terms). These "silicon samples" answered the same survey questions, and their answers were compared with those of real participants with the same backgrounds.

What they found. GPT-3's biases are not one uniform property but fine-grained and correlated with demographics, so suitable backstories make the model reproduce the response distributions of many human subgroups. The authors call this *algorithmic fidelity*, and argue that a model with enough of it can be a tool for studying people and society.

What it means here. Here the comparison with real people is made at the level of groups, not of one person as in

reading 3. It compares response distributions, which fits this part's advice to compare a group's spread with people's, not only its average. In the talk it appears on a backup page for Part 4, on whether a model can play the participant: it can pretest a study, not replace the sample.

What it does not show. The comparison covers American survey respondents on the questions studied. Later work finds that silicon samples can misrepresent and flatten identity groups, and that synthetic survey data can mislead (see How they connect).

5. Measuring and shaping a model's personality, psychometrically

Serapio-García et al., 2025, *Nature Machine Intelligence*

What they did. Established personality inventories were given to 18 language models under many prompt set-ups (different wordings and framings of the instructions), and the scores were examined for reliability and for validity against other measures, as for human tests. In reverse, instructions set traits to target levels, and the authors checked whether text generated afterwards expressed those levels.

What they found. For some models, under specific prompt set-ups, the measurements were reliable and valid, more clearly so for larger models and models further trained to follow instructions. A model's output could be shaped to mimic specific human personality profiles.

What it means here. A persona is a condition someone sets, so it needs a manipulation check: did it move the trait it names, only that trait, and under every wording that should mean the same? This study shows what such a check looks like with psychometrics applied in full. It is the check the talk points to on a backup page: each trait prompted on its own, from low to high, and mainly the trait that was prompted moved.

What it does not show. The questionnaire is the first check, and it is the one on the talk's backup page. A model told to be extraverted may score as extraverted by following the instruction rather than by having a trait, so behaviour that the persona did not name is a stronger test.

How they connect

Four threads run through these readings.

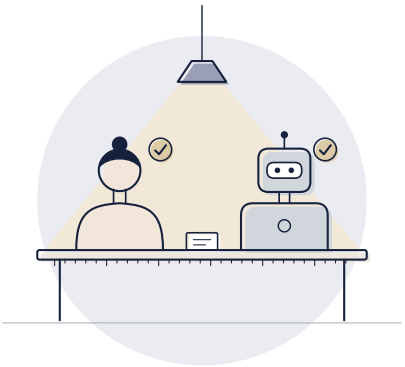
- **From test users to a town to real people.** Park et al. first let a model play the users of an online community that did not exist yet (2022), then built a town of agents (2023), then an agent for each of 1,052 real people, judged against the same person measured twice (2024). The line continues an older tradition of agent-based models of society, from Schelling's segregation model (1971, doi[20]) to Epstein's generative social science (1999, doi[21]1099-0526(199905/06)4:5%3C41::aid-cplx9%3E3.0.co;2-f)), now with agents that talk.
- **A persona is a manipulation.** Serapio-García et al. treat a model's personality as something to be measured with psychometric standards and shaped by instructions. A persona is therefore a condition the experimenter sets. Like any manipulation, it needs a check: did it move the trait it names, only that trait, under every wording that should mean the same?
- **Simulated samples and their critics.** Argyle et al. gave a model real respondents' backgrounds to create silicon samples. Later work found that such samples can mislead (Bisbee et al., 2024, doi[22]) and can misportray and flatten identity groups (Wang et al., 2025, doi[23]).
- **Classic paradigms as tests for agents.** Weng et al. took Asch's conformity design (1955, doi[24]) and ran it on language-model agents, with its controls; the talk keeps it on a backup page for questions.

The open question. Acting like a kind of person is not yet being like one. A simulated participant behaves as it does partly because of the persona someone gave it. To learn something about *real* people, first check that the persona did what it should, then compare the simulation with real people at the level the claim is about, such as individuals, groups or a whole classroom. Controls such as shuffled profiles and blind rating help, but they show what an output looks like, not what lies behind it.

[20] https://doi.org/10.1080/0022250x.1971.9989794
[21] https://doi.org/10.1002/(sici
[22] https://doi.org/10.1017/pan.2024.5
[23] https://doi.org/10.1038/s42256-025-00986-z
[24] https://doi.org/10.1038/scientificamerican1155-31

Part 4 · AI in your own research

How can you use it? Check it first, then use it.

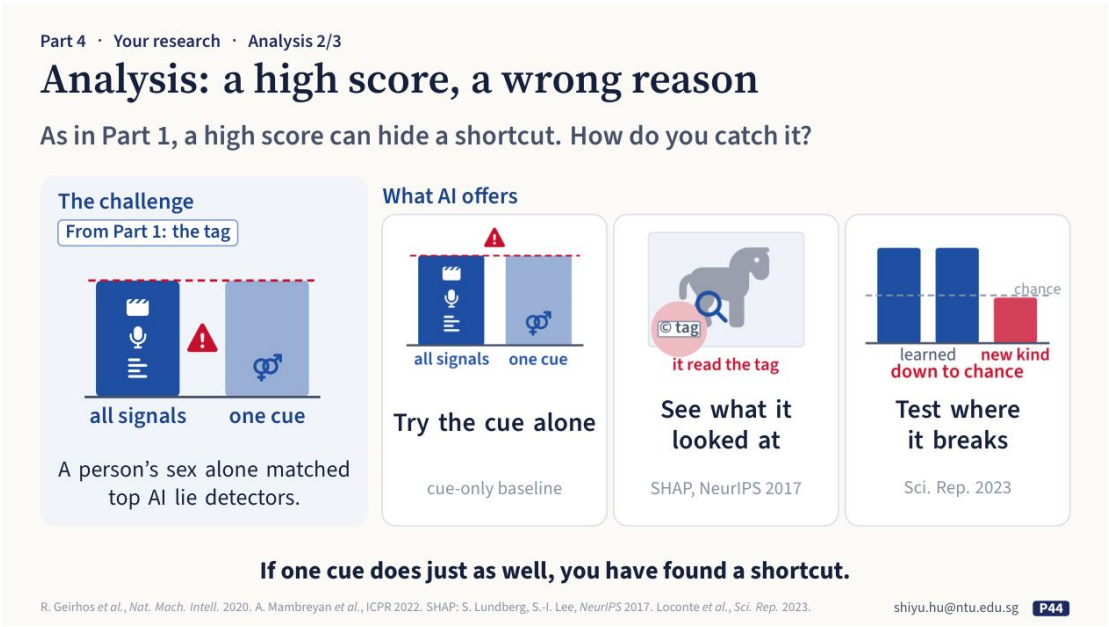


What the talk showed

The talk gives advice here rather than new studies, along the two ways of using AI named in its title. Parts 1 and 2 were analysis: a model reads behaviour, like a new measuring tool. Part 3 was modelling: a model takes a person's place. Each page starts from a challenge the talk has already met, then gives three things to try.

Analysis: few people, many trials. Clever Hans scored 98% when the questioner knew the answer and only 8% when the questioner did not, so a score depends on how you test. The same goes for a model: if one person is in both training and testing, the model has already seen them, and the score looks too good. With few people, do not start from zero. Take a model that has already learned from a large amount of data, such as LaBraM, pre-trained on about 2,500 hours of EEG (Jiang et al., ICLR 2024), and adjust it a little for your study. For a new person, some models can adjust from that person's unlabelled recordings alone, such as SPDIM (Li et al., ICLR 2025). And leave one person out: train on everyone else, test on that person, then do the same for the next. **Test on people the model has never seen.**

Analysis: a high score, a wrong reason. In Part 1 a model said *horse* because it was reading a source tag on the photos, and your field has its own tags. On one public data set, predicting only a person's sex did about as well as leading AI lie detectors (Mambreyan et al., 2022, accepted version[25]). So try the suspected cue alone. Ask what the model really looked at: tools such as SHAP (Lundberg & Lee, NeurIPS 2017) show which parts of the input mattered. And test it where the cue stops working: lie detectors trained on two kinds of lie fell to chance on a third (Loconte et al., 2023, doi[26]). In psychology's words, that is validity: the score is high, but it is not measuring what you meant. **If one cue does just as well, you have found a shortcut.**



Slide 44 of the talk: a high score, a wrong reason.

[25] https://www.repository.cam.ac.uk/handle/1810/338298
[26] https://doi.org/10.1038/s41598-023-50214-0

Analysis: people disagree. When a boy fell in Part 2, one person said he was pretending, one said crying and one said smiling. Your raters do this too. An average loses it, and so does a model trained on the average. So keep every rating, and let the model learn how people split, with soft labels. Then compare it with all your raters, as FIOVA did in Part 2. And know the limit, the noise ceiling: if your raters agree only so much with each other, do not expect more from a model. With something we cannot see directly, like well-being, a model trained on your scale learns the scale, not the thing itself. **Keep the disagreement: it tells you how far a model can go.**

Modelling: the model as a stand-in. A stand-in takes your participants' place: you try your questions on it before you recruit anyone, or it does your task, like a conformity test (Weng et al., 2025). The digital twins in Part 3 came from real interviews and were checked against the same people, and a stand-in needs that too. Otherwise the average can look right while everyone sounds too much alike (Bisbee et al., 2024, doi[27]). So build simulated people from real data, as silicon samples do. Look at the whole range of answers, not only the average (Meister et al., 2025, arXiv:2411.05403[28]). And check whether it has seen your test already: give it the first half of an item, and if it finishes it word for word, it has (Golchin et al., ICLR 2024). **Get the spread right, not only the average, before you trust a stand-in.**

Modelling: the model as a partner. A partner talks with your participants again and again. In Smallville, memory made the agents feel like people. A partner needs it even more, because it can forget what someone told it, or wander off the script. Give it a memory it can look things up in, and test it: tell it something in session one, and ask about it in session three (LoCoMo; Maharana et al., 2024, arXiv:2402.17753[29]). Then split the job: rules decide what it does, and the model only picks the words. Psychology already has a name for this, treatment fidelity (Bellg et al., 2004, doi[30]). And stay with one version: a model can change without telling you, and every participant should meet the same partner. An open model you can keep, such as OLMo (Groeneveld et al., ACL 2024), makes that possible. **Fix what it decides, let the words vary, and test its memory.**

On the backup pages. More is kept for the questions after the talk.

Many signals, out of step. Video, voice and brain signals each have their own clock, and one can disappear when a face turns away or a sensor falls off. Put everything on one clock by marking the time on every data point. In training, hide one signal at random, so that the model learns to manage without it. Afterwards, take away one signal at a time to see which one carries the result.

Three old traps, in a new form. Whatever you use a model for, three traps come back, and psychology knows each of them already. The model changes, like a measuring device that drifts: on the same prime-number questions, GPT-4 was right 84% of the time in March 2023 and 51% in June (Chen, Zaharia & Zou, 2024, doi[31]), so pin the version and date, and rerun a fixed test set. The wording moves the answer, as survey researchers have long known (Reiss, 2023), so try several wordings and report them all. And a classic task may test less than it seems, like a practised participant: GPT-3 solved a well-known card task but failed once the cards were reordered (Binz & Schulz, 2023, doi[32]), so compare old, reworded and new items.

Drafting, running and pretesting. One team let a model draft items for a new scale and for short Big Five scales, then tested them with more than 1,200 people (Götz et al., 2024, doi[33]). More than 2,000 people who believed a conspiracy theory talked with GPT-4 Turbo, and their belief fell by about 20%, still lower two months later (Costello et al., 2024, doi[34]; an erratum appeared on 22 October 2026, doi[35]). There the model is the manipulation, so record and read every conversation. A simulated participant can pilot your materials, but it is a pilot, not the sample.

Five places, and a new coder. One page sets out five places a model can enter a study, each with a check psychology already uses: a pilot with people for drafted items, a real sample after a simulated pretest, reading every conversation when a model talks with participants, two human coders when it codes, and new items when it is the participant. Another treats a model as a new coder, as psychology treats any coder (Gilardi et al., 2023, reading 2). Its example is 100 video descriptions, with each sentence labelled *seen* (what anyone can see) or *inferred* (a feeling or intention). Fix the prompt and settings, and run it three times; have two people label 30 descriptions and compare the model with each, using Cohen's kappa; reword the prompt once and see whether the labels hold; and report the model, version, date, prompt, runs and kappas. The reason behind the steps: a model that agrees with itself has not yet been shown to agree with people.

[27] https://doi.org/10.1017/pan.2024.5
[28] https://arxiv.org/abs/2411.05403
[29] https://arxiv.org/abs/2402.17753
[30] https://doi.org/10.1037/0278-6133.23.5.443
[31] https://doi.org/10.1162/99608f92.5317da47
[32] https://doi.org/10.1073/pnas.2218523120
[33] https://doi.org/10.1037/met0000540
[34] https://doi.org/10.1126/science.adq1814
[35] https://doi.org/10.1126/science.aem1178

The question behind it

How can you use it? Check it first, then use it. A model that reads your data, a simulated participant or an AI coder is a measurement tool you have not validated yet. Whichever way you use it, ask the questions of the first three parts: Right answer, right reason? (compare the mistakes, not only the scores); Whose standard does it follow? (compare with a group, not one person); Is it like us enough to study us? (compare it with real people). Record the model, version, date, full prompt, settings and number of runs, so that someone else can check.

Check it first, then use it: a high score is not enough, so compare the model with real people, and record the version and date.

What it gives back to psychology

Here the talk hands its checks back to psychology in psychology's own terms. A model in your study is one more tool to validate, and the validation is work psychology already knows: testing on new people, a control condition for the suspected cue, a real sample, treatment fidelity, two coders and an agreement statistic, new items for a familiar task. Used this way, a model can make some steps cheaper, such as drafting items, rehearsing a design or a first pass of coding, while the conclusions still rest on people.

After Part 4 and the summary, the talk closes with the speaker's own view, under the title "Why psychology and the humanities matter more now" (Why now). In some fields AI can now work like a PhD student, sometimes even like an expert, and every claim still needs checking. Making things gets cheap, and judging becomes scarce. So: outlast the model, by asking whether the work would still matter if models were ten times better; let AI do while you judge, and write the first draft yourself; and find your own crossing, by starting from a question your field asks best and bringing AI to it. When making things gets cheap, judging is the real work, and that has always been the job of psychology and the humanities.

Key readings

Read these five first. Each one below says what was done, what was found, what it means for this part's question, and what it does not show; its card gives the method in full and where the data are.

1. What language models can and cannot do for psychology

Demszky et al., 2023, *Nature Reviews Psychology*

What they did. Psychologists and computer scientists wrote this Perspective article together. It explains how language models produce human-like language without thinking or feeling like a person, then reviews their uses in psychological measurement, experimentation and practice.

What they found. The authors argue that models could advance all three, but are not yet ready for many of the most transformative applications. They examine four major concerns and recommend three investments: field-initiated "keystone" data sets, more standardised benchmarks, and shared computing and analysis infrastructure.

What it means here. Not shown in the talk. It is the broadest single map of the topic written for psychologists, and a good first reading: it covers all three uses in this part (a model as a coder, measuring constructs in text, a simulated participant) and the concerns that come with them. The other four readings each look at one use in detail.

What it does not show. It dates from 2023; the models have changed since, though the concerns have not gone away. It reviews and argues; it reports no new data.

2. A model as a coder: promising, and in need of validation

Gilardi et al., 2023, *PNAS*, read with Reiss, 2023

What they did. Gilardi and colleagues compared ChatGPT, given no examples ("zero-shot"), with crowd workers and trained annotators (what psychology calls coders) on four coding tasks in tweets and news articles (n = 6,183). Reiss repeated identical inputs on one real task, sorting website texts into news and not news, and varied the settings and the prompt wording.

What they found. ChatGPT's zero-shot accuracy beat that of crowd workers by about 25 percentage points on average, its intercoder agreement was higher than that of crowd workers and trained annotators on all tasks, and it cost about thirty times less. Reiss found that a small change in wording, or simply a repeat, can change the output; pooling repetitions helps.

What it means here. This is the point of the talk's backup page on coders: treat a model as a new coder, and check it

as psychology checks any coder, by agreement with others on the same material. Reiss shows one of the old traps in its new form: the wording moves the answer. Reading 4 takes up what follows once the labels are used.

What it does not show. Both studies cover particular tasks, languages and model versions at one moment; a result for one coding scheme does not transfer automatically to another.

3. Psychological text analysis in twelve languages

Rathje et al., 2024, *PNAS*

What they did. Three versions of GPT were given simple prompts, such as ”is this text negative?”, to detect sentiment, emotions, offensiveness and moral foundations in 47,925 tweets and news headlines in 12 languages that people had already annotated. Their answers were compared with the human annotations, dictionary methods and fine-tuned models (models trained further for one task).

What they found. GPT’s agreement with the human annotators ($r = 0.59$ to 0.77) was much higher than that of English-language dictionary analysis ($r = 0.20$ to 0.30), and close to, sometimes better than, several top fine-tuned models. Newer versions did better, especially for lesser-spoken languages.

What it means here. Not shown in the talk. This is the use psychologists are most likely to make of a model every day, measuring constructs in text, checked in a familiar way: against human judgement and the field’s existing methods. Because results changed from version to version, it also shows why the version must be recorded.

What it does not show. ”Accurate” here means agreement with the existing annotations, so the model is only as valid as they are. Correlations of 0.6 to 0.8 leave room for systematic differences on particular kinds of text.

4. Valid conclusions from AI labels and a few human labels

Angelopoulos et al., 2023, *Science*

What they did. This is a statistical framework, not an experiment. A small set of items labelled by people measures how far a model’s predictions are off, and that correction is applied to the estimate from the large set the model labelled. It gives confidence intervals for means, quantiles, and linear and logistic regression coefficients.

What they found. The intervals are provably valid, with no assumptions about the system that supplies the predictions; the more accurate the predictions, the narrower they are. Demonstrations come from six fields, from proteomics to ecology.

What it means here. Not shown in the talk. It answers the question that follows reading 2: if a model codes most of the data, what can still be concluded, and how many human labels are needed? It builds this part’s lesson for coders, validate against people every time, into the estimate itself.

What it does not show. The human-labelled items must be a random sample from the same population as the model-labelled ones. The method corrects the model’s errors on average; it does not make a biased model’s labels valid item by item.

5. Predicting the results of experiments before running them

Ashokkumar et al., 2026, *Nature*

What they did. The authors prompted a model to simulate how representative American samples would respond to 70 preregistered US survey experiments (469 effects, 119,330 participants), and compared the simulated responses across conditions. Fifteen megastudies (606 effects) served as a further test.

What they found. GPT-4’s predictions correlated strongly with the actual effects, about as accurately as pooled forecasts by people, including for studies not published or posted before its training data were collected (its training cutoff). The predictions systematically overestimated effect sizes. For the megastudies, correlations were lower but comparable to those of pooled expert forecasters.

What it means here. Not shown in the talk, it bears out the talk’s point that a simulated participant can pretest a study but does not replace the sample. It supports one careful use: predicting the direction of effects to choose, pilot or prioritise studies. It follows Part 3’s silicon samples; the check is Part 3’s: compare it with real people.

What it does not show. The experiments are American survey experiments. A good correlation across many effects does not make any single prediction reliable, and overestimated effect sizes would mislead a power calculation used in place of a pilot with people.

How they connect

Three threads run through these readings.

- **A model as a coder.** Psychology already knows how to check a coder: agreement with other coders on the same material. Gilardi et al. found a model more accurate and more consistent than crowd workers. Reiss found its output could change with the wording of the prompt or simply on a second run. Pangakis and Wolken (2025, *Keeping humans in the loop: Human-centered automated annotation with generative AI*, ICWSM, doi[36]) draw the conclusion: validate against people every time. Prediction-powered inference and design-based supervised learning (Egami et al., 2023, doi[37]) then show how to draw valid conclusions from model labels plus a set of human ones.
- **Measuring constructs in text.** Rathje et al. compare a model with the dictionary methods psychologists have long used, and find it much closer to human judgement, in many languages.
- **A simulated participant as a pretest.** Silicon samples (a key reading in Part 3; in the talk, on a backup page for this part) raised the idea; Ashokkumar et al. test it on 70 real experiments and find good predictions of direction and inflated effect sizes. Crockett and Messeri (2026, *AI surrogates and illusions of generalizability in cognitive science*, doi[38]) and Gao et al. (2025, *Take caution in using LLMs as human surrogates*, doi[39]) set out why a simulation should not take the place of the sample.

Demszky et al. cover all three uses, and the concerns that come with them.

The open question. Using a model in your research turns it into one of your measurement tools. The checks from the first three parts apply to it as to any tool: compare the mistakes, not only the scores (where does it go wrong, and does a small change in the prompt change the answer?); compare with a group, not one person (is it validated against more than one person?); and compare it with real people (a pretest, not the sample). Record the model, version, date, full prompt, settings and number of runs, so that someone else can check.

[36] https://doi.org/10.1609/icwsm.v19i1.35883
[37] https://doi.org/10.52202/075280-3000
[38] https://doi.org/10.1016/j.tics.2025.09.012
[39] https://doi.org/10.1073/pnas.2501660122

Glossary

One line per term; the full entries, with sources, are in docs/glossary.md.

Part 1 · One person: right answer, right reason?

- **Cue.** Any feature of an input that a person or a model can use to reach an answer, such as an object's shape, its texture, a watermark in the corner or the wording of a question. Two cues can lead to the same answer on ordinary items and to different answers on unusual ones.
- **Cue conflict.** A stimulus built so that two cues point to different answers, for example an image with the shape of a cat and the texture of elephant skin. The answer shows which cue won.
- **Shape bias.** On cue-conflict images, the share of decisions that follow the shape rather than the texture. People are strongly shape-biased; networks trained on ImageNet (a large set of labelled photographs) are texture-biased.
- **Error consistency.** How often two decision makers are wrong on the *same* items, beyond what their accuracies would produce by chance. It is computed as Cohen's κ on right and wrong, trial by trial.
- **Cohen's kappa.** An agreement statistic for two raters who sort items into categories. It corrects the observed agreement for the agreement expected by chance. $\kappa = 0$ is chance level and $\kappa = 1$ is perfect agreement.
- **Clever Hans effect.** A right answer produced by an unintended cue. The name comes from Clever Hans, a horse who seemed to count and calculate but was responding to tiny, unintended movements of the questioner's head. Oskar Pfungst showed this in 1904 by removing the cue.
- **Shortcut learning.** Learning decision rules that perform well on standard benchmarks (shared test sets) but fail to transfer to more challenging test conditions, such as real-world scenarios. This follows the definition of Geirhos et al. (2020).
- **Human baseline.** How people perform on the same items under the same conditions as the model. It is the reference against which "human-level" is judged.
- **Out-of-distribution data.** Test items that differ systematically from the training data, such as noisy, stylised or edge-only images for a model trained on photographs. The term is often shortened to OOD.
- **Psychophysics.** The measurement of how physical properties of a stimulus relate to perception, using controlled stimuli, many trials and careful presentation conditions.
- **Explanation method.** A technique that shows which parts of an input a model relied on for a decision, often as a heat map over an image. It is also called a relevance or attribution method.

Part 2 · A group: whose standard does it follow?

- **Reference.** The human data a model is compared with: one label per item, a majority vote, several raters, or the full distribution of people's answers.
- **Inter-rater reliability.** How far independent raters agree when they judge the same items.
- **Krippendorff's alpha.** An agreement coefficient for any number of raters, for nominal, ordinal, interval or ratio data, and for data with missing ratings. $\alpha = 1$ is perfect agreement and $\alpha = 0$ is chance level.
- **Gold label and majority vote.** A gold label is the single "correct" answer assigned to an item. A majority vote is the most common answer among several raters. Most data sets in AI store one of the two.
- **Label distribution.** The share of people who chose each answer for an item, for example 70% cat and 30% dog. A *soft label* is this distribution used as the training target, instead of one answer.
- **Human label variation.** The fact that people give different, plausible answers to the same item, and the research that studies it instead of treating it as error.
- **Perspectivism.** The view that, for many tasks, there is no single ground truth, and that the perspectives of different groups of raters should be kept and modelled.
- **Seen and inferred.** The distinction between what is observable in an input, such as an action or an event, and what is read into it, such as an intention or a feeling. People usually agree more on the first than on the second.
- **Rater model.** A statistical model of how each rater produces labels, for example each rater's error rates, used to combine many imperfect ratings.

- **Learning from human feedback.** Training a model on people's judgements of its outputs, typically by having people compare or rank answers, training a reward model on those comparisons, and tuning the model to score well on it.
- **LLM as a judge.** Using a large language model to grade or compare answers, in place of human judges.

Part 3 · A society: like us enough to study us?

- **Machine playing people.** A model set up to play people: a virtual student, a simulated class or town, a group of agents, or a participant in a study. Sometimes it plays the people around someone, such as the class a trainee teacher faces; sometimes it answers in a participant's place. Either way, the conclusion drawn is about how real people behave.
- **Persona.** A description given to a model, in its instructions, of who it should be: traits, background, opinions or a role.
- **Manipulation check.** A test of whether a manipulation did what it was meant to do. For a persona, the talk's check: did it move the trait it names, and mainly that trait? Wordings that should mean the same should also give the same result.
- **Silicon sample.** A set of model responses produced by conditioning a language model on the backgrounds of real survey respondents, used in place of a human sample. The term comes from Argyle et al. (2023).
- **Algorithmic fidelity.** The degree to which a model, suitably conditioned, reproduces the response distributions of particular human subgroups. The term comes from Argyle et al. (2023).
- **Turing test.** Turing's imitation game: a judge chats by text with a machine and a human and says which is which. In the standard three-party version, the judge talks to both at once.
- **Generative agent.** A software agent built on a language model, with memory, reflection and planning, that produces believable behaviour over time and interacts with other agents.
- **Conformity.** Changing one's answer to match a group. Asch showed that many people give a clearly wrong answer when everyone before them does.
- **Agent-based model.** A simulation in which many individual agents follow behavioural rules and interact, so that group-level patterns emerge from individual behaviour.

Part 4 · AI in your own research

- **AI coder.** A language model used to label or code material for research, such as sorting texts into categories, rating sentiment or coding open-ended answers.
- **Prompt sensitivity.** The change in a model's output when the wording of the instruction changes in ways that should not matter, or when the same input is simply run again.
- **Prediction-powered inference.** A statistical framework that combines a large set of model predictions with a smaller set of human labels to give valid confidence intervals, without assuming the model is correct.
- **Simulated participant.** A language model prompted to respond as a research participant would, used to pretest materials, predict results or, controversially, replace a sample.

Working together: psychology and AI research

This page is written by an AI researcher for psychology researchers and students who would like to work with AI researchers on the question of the talk: **it looks capable; how would we know it really is?** Neither side can answer it alone. AI research has the models, the benchmarks (standard test sets) and the engineering to run them on many items. Psychology has the methods that tell a capable-looking performance from a capable one: controlled designs, measurement and validity, samples of real people, reliability and ethics.

The page sets out what each side brings, where the two fields meet (one place for each of the talk's four questions), what to agree on before you start, and the misunderstandings that come up most often in both directions. Each place comes with readings and published studies to learn from.

What each side brings

Area	Psychology brings	AI research brings
The question	a construct, and what would count as evidence for it	a model, and the task it is said to do well
Design	paradigms, control conditions, counterbalancing, pre-registration	access to models, prompts and settings; running thousands of items, many times
People	samples, recruitment, consent and ethics approval	benchmarks and data sets, often with one label per item
Measurement	reliability, validity, psychometrics	metrics such as accuracy, and item-level outputs
Analysis	effect sizes, comparison with the spread among people	explanation methods, training and fine-tuning (further training of an existing model), simulations of many agents

Where the two fields meet

1. Right answer, right reason?

Part 1 · One person. The check: compare the mistakes, not only the scores.

- **The question.** A model scores close to people on a task. Does it get there for the same reasons?
- **Psychology brings.** The task and its items; the cue you suspect the model uses; a control condition that separates that cue from the one you meant to test; a sample of people tested under the same conditions.
- **AI research brings.** One or several models run on the same items, with item-level output; for image models, explanation maps that show where a model looked.
- **Read first.** Geirhos, Meding & Wichmann (2020) on error consistency; Pfungst (1911) on Clever Hans.
- **A study to learn from.** Close scores, different mistakes.

2. Whose standard does it follow?

Part 2 · A group. The check: compare with a group, not one person.

- **The question.** A model is said to agree with "the human label". Whose label is it, and how much do people agree with each other?
- **Psychology brings.** A rating design; how raters are sampled; inter-rater reliability; keeping what is seen (an action) apart from what is inferred (a feeling).
- **AI research brings.** Model outputs on many items and repeated runs; evaluation, or training, against the distribution of people's answers rather than a single label.
- **Read first.** Heider & Simmel (1944); Morgenroth et al. (2025), Emo-FilM, *Scientific Data*, doi[40]. In Emo-FilM, raters agreed only moderately on emotion moment by moment.
- **Studies to learn from.** About fifty people per image and Forty-four raters, fourteen films.

[40] https://doi.org/10.1038/s41597-025-04803-5

3. Is it like us enough to study us?

Part 3 · A society. The check: compare it with real people.

- **The question.** A simulated participant, or a group of agents, acts like a kind of person. Does the persona do what it says, and does the simulation match real people?
- **AI research brings.** Personas, agents and multi-agent simulations; running them many times, under several wordings of the same instruction.
- **Psychology brings.** The manipulation check; validated questionnaires and behavioural measures; data from real people to compare with; the blind rating.
- **Read first.** Serapio-García et al. (2025); Turing (1950); Jones & Bergen (2026); Park et al. (2023).
- **Studies to learn from.** Measuring and shaping a model's personality and A three-party Turing test, with a persona.

4. How can you use it?

Part 4 · AI in your own research. The check: check it first, then use it.

- **The question.** A model reads your data, or takes a person's place: it codes your material, pretests your study or talks with participants. What can you still conclude about people?
- **AI research brings.** Choosing a model and setting it up; running it the same way every time; keeping the outputs of every run.
- **Psychology brings.** The coding scheme and the human coders the model is checked against; new people to test it on, and a control for the suspected cue; the real sample, of which a simulated pretest is no replacement; the validation.
- **Studies to learn from.** A model as a coder and Predicting the results of experiments.

Before you start: what to agree on

- **The claim**, in one sentence, and the level it is about: one person, a group or a society.
- **The data.** Who provides which data; public data where possible; ethics approval and consent before anything is collected from people.
- **The model record.** The model, version, date, full prompt, settings and number of runs, kept for every condition.
- **What counts as a pass**, decided before anyone looks.
- **Who does what**, and how authorship will be decided.

Common misunderstandings, in both directions

- **"A higher score means more human-like."** Close scores can hide different mistakes; compare which items each gets wrong (Part 1).
- **"The human label is the answer."** It is one reading of the item, and people often disagree; compare the model with the spread among people (Part 2).
- **"If people cannot tell it apart, it is like people."** A blind rating shows that the output looks human, not what lies behind it (Part 3).
- **"One prompt is one experiment."** A model's output can change with the wording of the prompt, or simply on a second run; reword, repeat and report every run (Part 4).
- **"A model is a black box, so its behaviour cannot be studied."** Its behaviour can be studied with the designs psychology already uses for people: controlled items, item-level data and manipulation checks.
- **"A simulated sample can replace the participants."** Simulated samples have predicted the direction of real effects but overestimated their size; use them as a pretest, not as the sample (Part 4).

Start where psychology is strongest: measurement and design.

Read more

In the repository: `resources/reading-path.md` (a step-by-step reading path), `resources/examples/` (one card per key reading), `resources/students.md` (reading list by area of psychology), `resources/README.md` (by part of the talk) and `resources/literature.md` (about 300 references, field-wide).

Text and diagrams: CC BY-NC 4.0. The talk's drawings: © 2026 Shiyu Hu, all rights reserved. Built from the repository by `docs/handout/make_handout.py`.